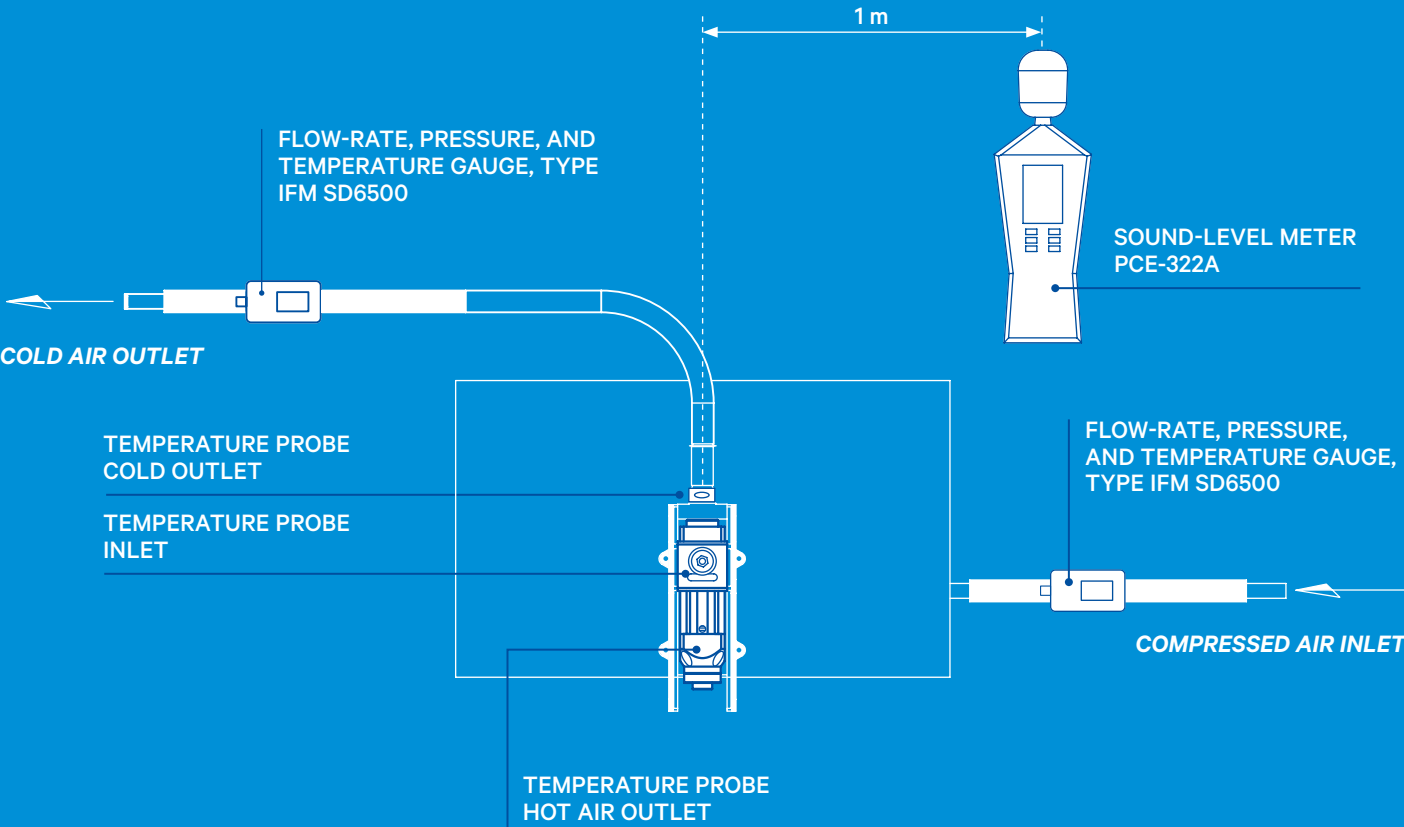
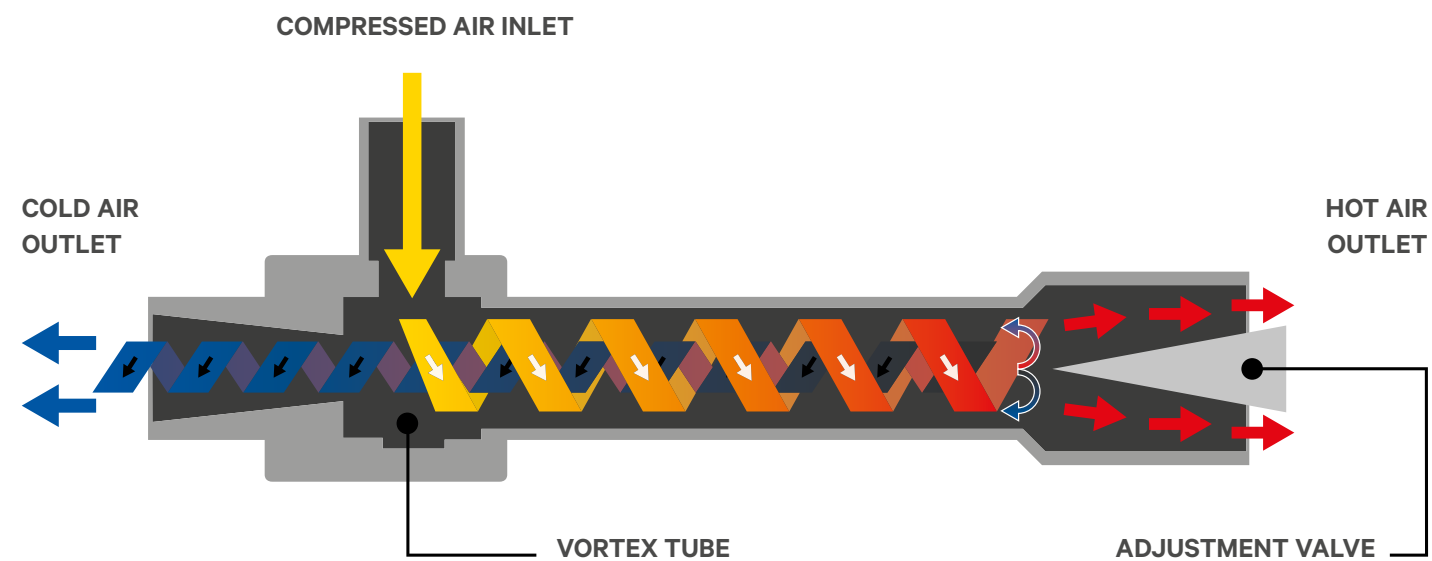


Description of the set-up and instrumentation used for our tests on pneumatic coolers.



The **VR / VRX / VR U-G SERIES** coolers are state-of-the-art solutions for compressed-air cooling based on the principle of the Vortex Tube. The excellent performances of flow-rate and ΔT generated, the design, the fastenings that make them extremely versatile to mount, and the possibility to combine them in a patented system with the air amplifiers (to use the hot air flow), offer customers an innovative, effective, and inexpensive solution to cool down metal and plastic parts, electric and electronic control cabinets, and mechanical applications. All this with a simple connection to the compressed-air line.

- ΔT up to -40°C for the cold flow and $+60^{\circ}\text{C}$ for the hot flow, in comparison to the temperature of air at inlet
- Easy to install, thanks to flanges and magnetic supports
- Patented system of hot air's recovery to actuate an amplifier/conveyor
- Made of corrosion-resistant materials
- No moving part, so not subject to wear and tear
- No electricity or chemical substances required
- They do not cause either sparkles or interferences
- Instant operation
- Reliable and maintenance-free



Ranque-Hilsch tube (Vortex tube)

DESCRIPTION OF VORTEX TUBES

The Ranque-Hilsch tube, in the industrial sector better known as “Vortex tube”, is a device that splits a compressed-air flow in 2 separate streams: one of cold air, and one of hot air.

The core of the system is the vortex chamber, which is connected to 2 opposed tubes, one of which features a valve. When the compressed air is injected tangentially in the chamber, this causes the rotary movement of air towards one of the exits. This vortex moves rotating at high speed and brushing against the inner side of the tube, increasing in temperature; the valve placed at the hot air outlet enables some of it to be exhausted. The remaining part goes back, creating a low pressure vortex moving towards the other exit and giving away heat to the first vortex. So, this flow is much colder.

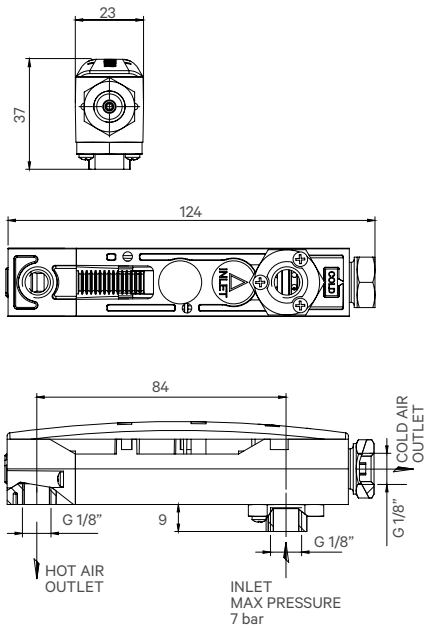
The ΔT generated is inversely proportional to the volume of the flow. The differences in temperature are considerable and can reach -40°C for the cold flow and 60°C for the hot flow.

In the industrial field the Vortex tubes have been employed for a long time and have found a variety of applications in which they offer a major added value. They have great cooling performances, are very easy to install and have instant operation, have no moving part and therefore are maintenance-free. Plus, they do not require electric power, so they are suitable for dangerous environments and humid areas.

If the application enables their use, they are price-worthier than electric coolers. Our coolers SERIES VR and VRX, beside the excellent performances in comparison to the other products in the market, were designed to be easily customised according to customers' demands.

SERIES VR-100

MODULAR PNEUMATIC COOLERS



GENERAL FEATURES - VR-100

Materials	Body and cover: Nylon 6.6
	Air connections and nozzles: brass
Air inlet port	G-1/8" F
Outlet port (cold flow)	G-1/8" F
Exhaust port (hot flow)	G-1/8" F
Recommended hose	Ø-8x1
Air supply pressure	3 ÷ 7 bar
Cooling power*	120 W - 100 Kcal/h - 400 BTUH
Optional magnetic support	KACM-VR100

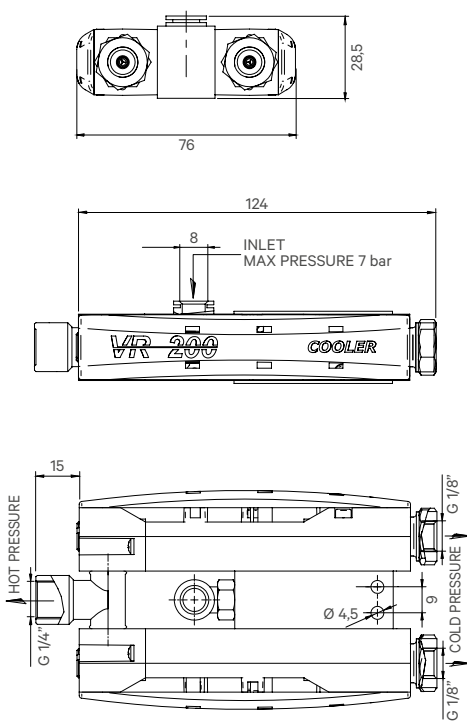
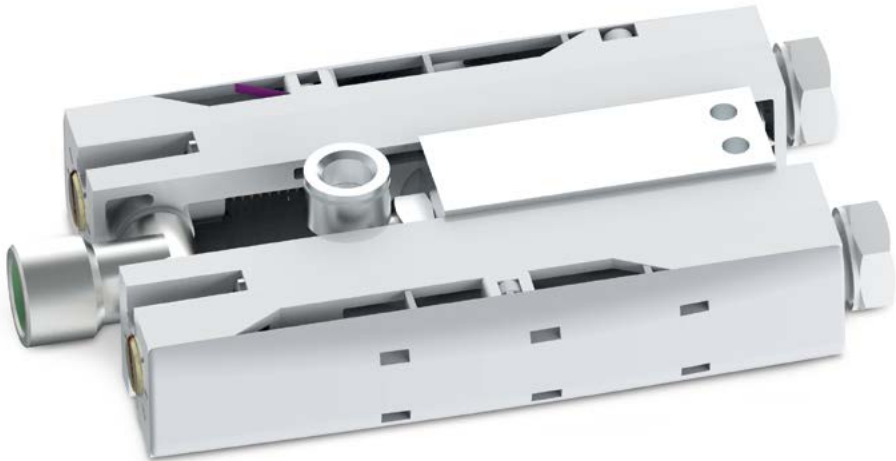
*with inlet pressure 7 Bar and inlet temperature 20°C

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min
1	-1.5	32
2	-8	53
3	-15	74
4	-21,5	94
5	-24,5	115
6	-26,5	135
7	-28	154

SERIES VR-200

MODULAR PNEUMATIC COOLERS



GENERAL FEATURES - VR-200

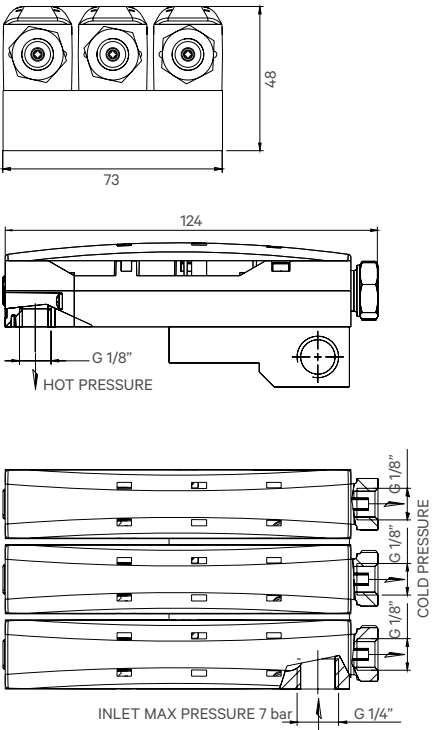
Materials	Body and cover: Nylon 6.6
	Air connections and nozzles: brass
Air inlet port	Push-in fitting Ø-8x6
Outlet port (cold flow)	2 x G-1/8" F
Exhaust port (hot flow)	2 x G-1/8" F
Recommended hose	Ø-8x1
Air supply pressure	3 ÷ 7 bar
Cooling power*	240 W - 200 Kcal/h - 800 BTUH
Optional magnetic support	KACM-VR200

*with inlet pressure 7 Bar and inlet temperature 20°C

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min
1	-1,5	64
2	-8	106
3	-15	148
4	-21,5	196
5	-24,5	230
6	-26,5	270
7	-28	308

SERIES VR-300T • 3 OUTLETS
MODULAR PNEUMATIC COOLERS



GENERAL FEATURES - VR-300T

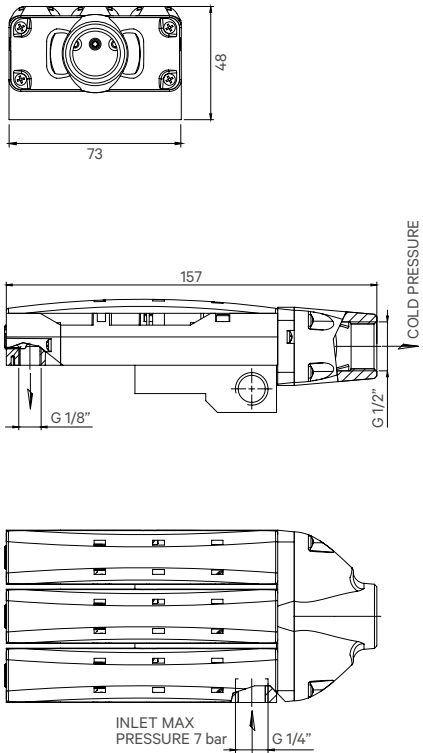
Materials	Body and cover: Nylon 6.6
	Air connections and nozzles: brass
Air inlet port	G-1/4" F
Outlet port (cold flow)	3 x G-1/8" F
Exhaust port (hot flow)	3 x G-1/8" F
Recommended hose	Ø-8x1
Air supply pressure	1 ÷ 7 bar
Cooling power*	360 W - 300 Kcal/h - 1200 BTUH
Optional magnetic support	KACM-VR300

*with inlet pressure 7 Bar and inlet temperature 20°C

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min
1	-1,5	96
2	-8	159
3	-15	222
4	-21,5	282
5	-24,5	345
6	-26,5	405
7	-28	462

SERIES VR-300U • SINGLE OUTLET
MODULAR PNEUMATIC COOLERS



GENERAL FEATURES - VR-300U

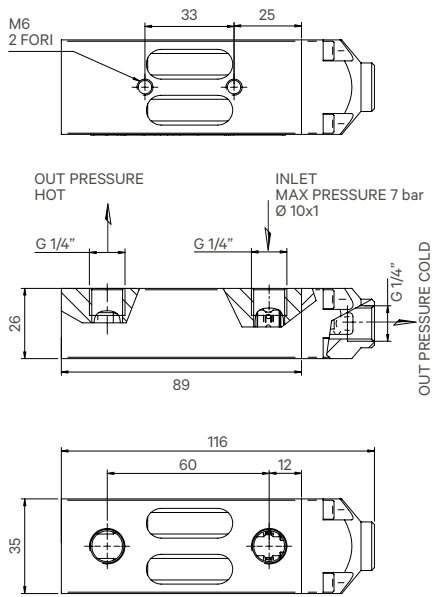
Materials	Body and cover: Nylon 6.6
	Air connections and nozzles: brass
Air inlet port	G-1/4" F
Outlet port (cold flow)	1 x G-1/2" F
Exhaust port (hot flow)	3 x G-1/8" F
Recommended hose	Ø-8x1
Air supply pressure	3 ÷ 7 bar
Cooling power*	360 W - 300 Kcal/h - 1200 BTUH
Optional magnetic support	KACM-VR300

*with inlet pressure 7 Bar and inlet temperature 20°C

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min
1	-1,5	96
2	-8	159
3	-15	222
4	-21,5	282
5	-24,5	345
6	-26,5	405
7	-28	462

SERIES VR-200U • SINGLE OUTLET
HIGH-PERFORMANCE COMPACT PNEUMATIC COOLERS



GENERAL FEATURES - VR-200U	
Materials	Body and cover: Delrin Ports and nozzles: Brass
Air inlet port	G-1/4" F
Outlet port (cold flow)	G-1/4" F
Exhaust port (hot flow)	G-1/4" F
Recommended hose	Ø-8x1
Air supply pressure	max 7 bar
Cooling power*	264 W - 220 Kcal/h - 880 BTUH
Optional magnetic support	By means of 2 threads M6 on the body
Weight	210 g

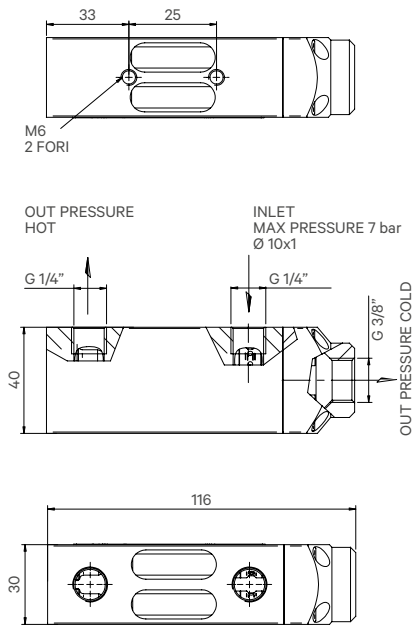
* With inlet pressure 7 Bar and inlet temperature 20°C.

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min	Noise level* dBA
1	-2	64	50
2	-12	106	54
3	-18	148	58
4	-23	188	61
5	-26	230	65
6	-28	270	68
7	-31	308	70

*Test made with insulated LOC-LINE flexible tube at cold outlet, and tube L= 1 m at hot air outlet.

SERIES VR-400U • SINGLE OUTLET
HIGH-PERFORMANCE COMPACT PNEUMATIC COOLERS



GENERAL FEATURES - VR-400U	
Materials	Body and cover: Delrin Ports and nozzles: Brass
Air inlet port	G-1/4" F
Exhaust port (hot flow)	G-1/4" F
Recommended hose	Ø-10x1
Supply pressure	max 7 bar
Cooling power*	528 W - 440 Kcal/h - 1760 BTUH
Optional fixation of magnetic kit	By means of 2 threads M6 on body
Weight	285 g

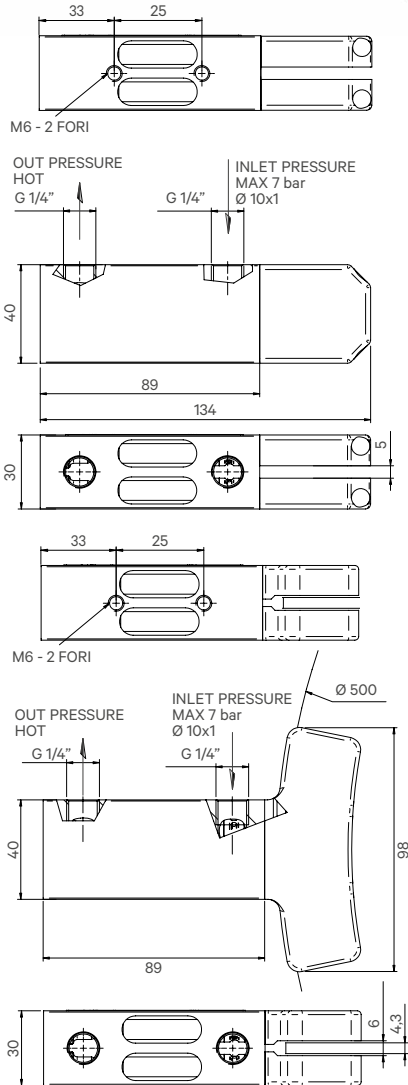
* With inlet pressure 7 Bar and inlet temperature 20°C.

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min	Noise level* dBA
1	-2	128	55
2	-12	212	60
3	-18	296	64
4	-23	376	67
5	-26	460	70
6	-28	540	71
7	-31	616	73

*Test made with insulated LOC-LINE flexible tube at cold outlet, and tube L= 1 m at hot air outlet.

SERIES VR-400G • TO COOL DOWN BLADES, BELTS, AND BANDS
HIGH-PERFORMANCE COMPACT PNEUMATIC COOLERS



GENERAL FEATURES - VR-400G

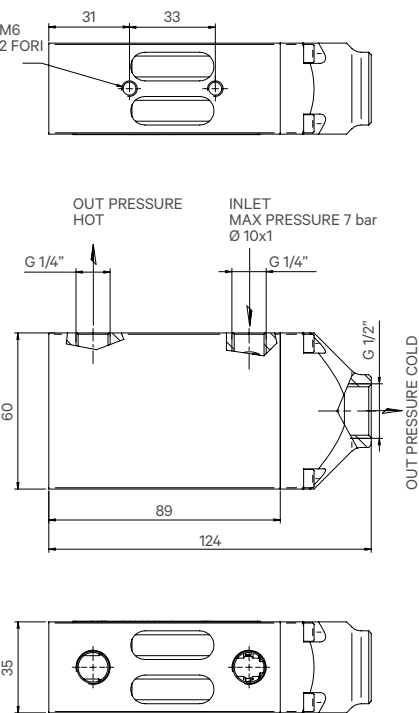
Materials	Body: Derlin Clamps: ABS (other materials on request) Inner spindles: brass
Air inlet port	G-1/4" F
Clamps width (cold flow)	5 mm (customised dimensions on request)
Outlet port (hot flow)	G-1/4" F
Recommended hose	Ø-10x1
Air supply pressure	max 7 bar
Cooling power*	528 W - 440 Kcal/h - 1760 BTUH
Fixation	By means of two M6 threads on body
Weight	340 g

* With inlet pressure 7 Bar and inlet temperature 20°C..

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min
1	-2	128
2	-12	212
3	-18	296
4	-23	376
5	-26	460
6	-28	540
7	-31	616

SERIES VR-600U • SINGLE OUTLET
HIGH-PERFORMANCE COMPACT PNEUMATIC COOLERS



GENERAL FEATURES - VR-600U

Materials	Body: Derlin Ports and nozzles: Brass
Air inlet port	G-1/4" F
Clamps width (cold flow)	G-1/2" F
Outlet port (hot flow)	G-1/4" F
Recommended hose	Ø-10x1
Air supply pressure	max 7 bar
Cooling power*	720 W - 600 Kcal/h - 2400 BTUH
Fixation	By means of two M6 threads on body
Weight	460 g

* With inlet pressure 7 Bar and inlet temperature 20°C..

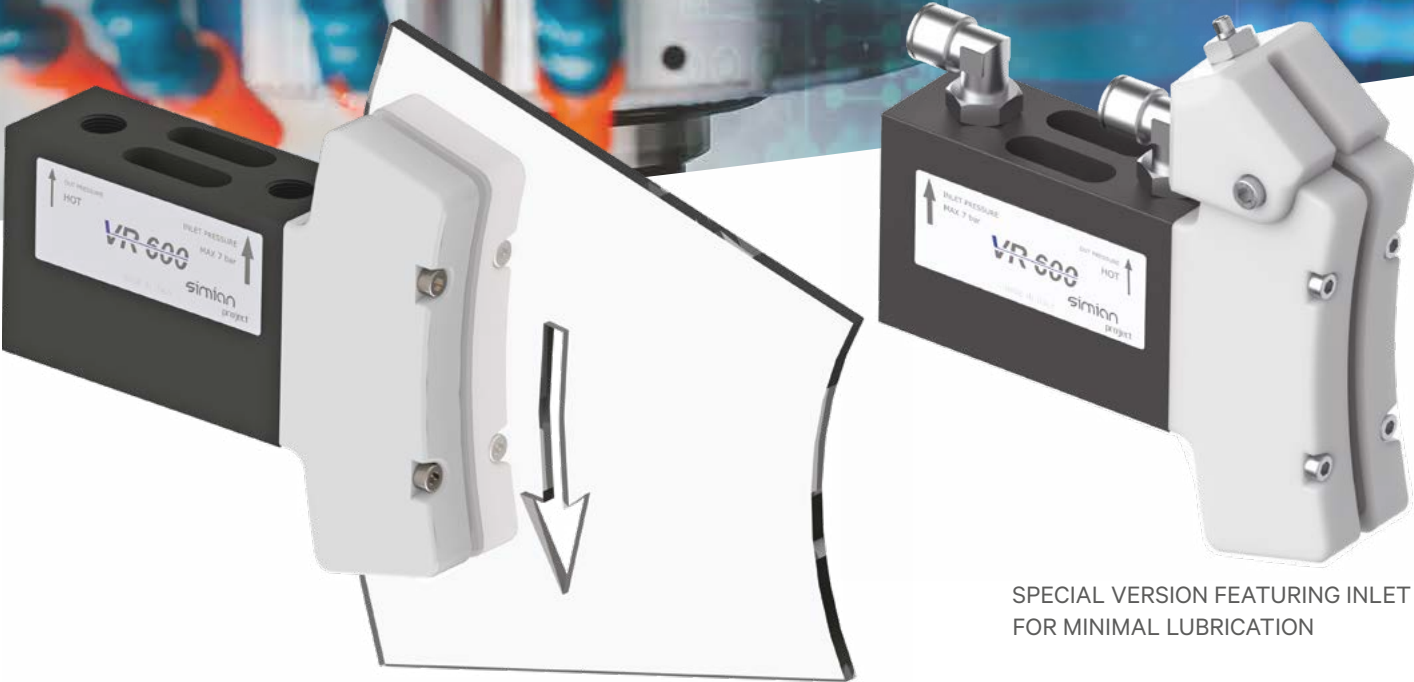
PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min	Noise level* dBA
1	-2	192	58
2	-12	318	64
3	-18	444	68
4	-23	564	72
5	-26	690	75
6	-28	810	78
7	-31	924	80

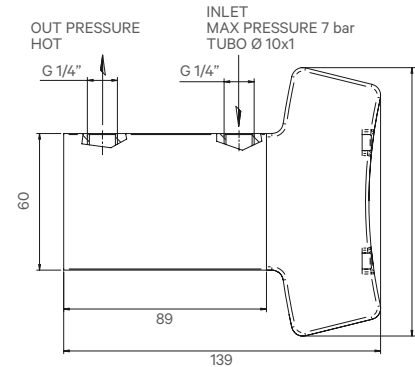
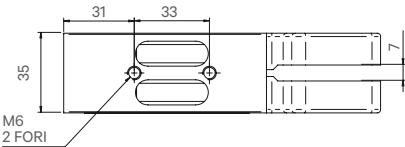
*Test made with insulated LOC-LINE flexible tube at cold outlet, and tube L= 1 m at hot air outlet.

SERIES VR-600G • TO COOL DOWN BLADES, BELTS, AND BANDS

HIGH-PERFORMANCE COMPACT PNEUMATIC COOLERS



SPECIAL VERSION FEATURING INLET FOR MINIMAL LUBRICATION



GENERAL FEATURES- VR-600G

Materials	Body: Derlin
	Clamps: ABS (other materials on request)
	Inner spindles: Brass
Air inlet port	G-1/4" F
Clamps width (cold flow)	11 mm (customised dimensions on request)
Outlet port (hot flow)	G-1/4" F
Recommended hose	Ø-10x1
Air supply pressure	max 7 bar
Cooling power*	790 W - 660 Kcal/h - 2640 BTUH
Fixation	By means of two M6 threads on body
Weight	540 g

* With inlet pressure 7 Bar and inlet temperature 20°C..

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min
1	-2	192
2	-12	318
3	-18	444
4	-23	564
5	-26	690
6	-28	810
7	-31	924